Dataset	Head	Accuracy	Macro-F1	Acc. 95% CI
Fashion-MNIST	LASTLAYER	0.8829 ± 0.0063	0.8826 ± 0.0066	[0.8752, 0.8907]
	CONCAT	0.8789 ± 0.0020	0.8783 ± 0.0020	[0.8764, 0.8814]
	SCALARMIX	0.8807 ± 0.0026	0.8800 ± 0.0027	[0.8775, 0.8839]
	LAYA	0.8834 ± 0.0046	0.8827 ± 0.0052	[0.8777, 0.8891]
CIFAR-10	LASTLAYER	0.7112 ± 0.0236	0.7108 ± 0.0228	[0.6819, 0.7405]
	CONCAT	0.7155 ± 0.0096	0.7136 ± 0.0117	[0.7036, 0.7274]
	SCALARMIX	0.7144 ± 0.0171	0.7123 ± 0.0177	[0.6932, 0.7356]
	LAYA	0.7212 ± 0.0060	0.7205 ± 0.0053	[0.7138, 0.7286]
IMDB	LASTLAYER	0.8692 ± 0.0057	0.8691 ± 0.0057	[0.8621, 0.8762]
	CONCAT	0.8688 ± 0.0062	0.8687 ± 0.0063	[0.8611, 0.8765]
	SCALARMIX	0.8669 ± 0.0043	0.8669 ± 0.0044	[0.8614, 0.8723]
	LAYA	0.8707 ± 0.0035	0.8707 ± 0.0035	[0.8663, 0.8751]

Table 1: Quantitative comparison of output heads across datasets (5 runs per configuration). All models shared the same backbone and were trained under identical conditions. No data augmentation or task-specific optimization was applied.

Dataset	d^*	τ	$\psi(\cdot)$	Scorer Width
Fashion-MNIST	96	1.5	`identity`	192
CIFAR-10	128	1.0	`identity`	128
IMDB	96	1.3	`mlp`	192

Table 2: Best-performing LAYA configurations selected by grid search on each dataset.

Fashion-MNIST, CIFAR-10, and IMDB. For every dataset, the mean and standard deviation of $\alpha_i(x)$ were calculated, providing a compact summary of the average contribution of each layer to the final prediction (Fig. 1). The resulting profiles show that attention does not collapse to a uniform distribution: LAYA exhibits clear, dataset-specific depth preferences. On Fashion-MNIST, attention is primarily concentrated on the deepest layer (mean ≈ 0.76), although shallower layers still display notable variability, indicating that different samples rely on different depths even in a relatively simple visual task. On CIFAR-10, the distribution becomes markedly sharper, with the final layer dominating almost entirely (mean ≈ 0.96, with low variance), reflecting the higher representational demands of natural-image classification. The IMDB sentiment classifier shows a contrasting pattern: attention is evenly split between the two recurrent layers (means ≈ 0.48 and ≈ 0.52, both with slight variance), suggesting that the BiLSTM stack preserves complementary information across depths and that both layers contribute consistently to the decision.

These results indicate that LAYA adapts its depth usage in a strongly task-dependent manner: concentrated on deeper representations in complex vision tasks, more distributed in simpler visual settings, and balanced in text. This behavior confirms that LAYA does not merely aggregate layers but actively reweights representational depth in alignment with the structure and requirements of each domain.

More importantly, the global attention statistics provide a direct diagnostic tool for understanding how representational importance is allocated across the network. By quantifying the average contribution of each layer, LAYA enables identification of depth levels that consistently carry little predictive weight, detection of shifts in depth usage across tasks or modalities, and characterization of how architectural components participate in the decision process. Such information would otherwise require external probing methods or post hoc explanations.

Class-wise attention profiles. To investigate whether different semantic categories relied on different abstraction levels, class-wise attention profiles were computed for each dataset. For every class c and layer i, attention coefficients were averaged over all test samples with true label c,

$$\bar{\alpha}_{c,i} = \mathbb{E}[\alpha_i(x) \mid y = c],$$

and the analysis was further stratified into *all*, *correct*, and *incorrect* predictions. The resulting heatmaps (Fig. 2) provide a compact view of how LAYA distributes importance across depth for each class and allow a direct comparison across the three benchmarks.

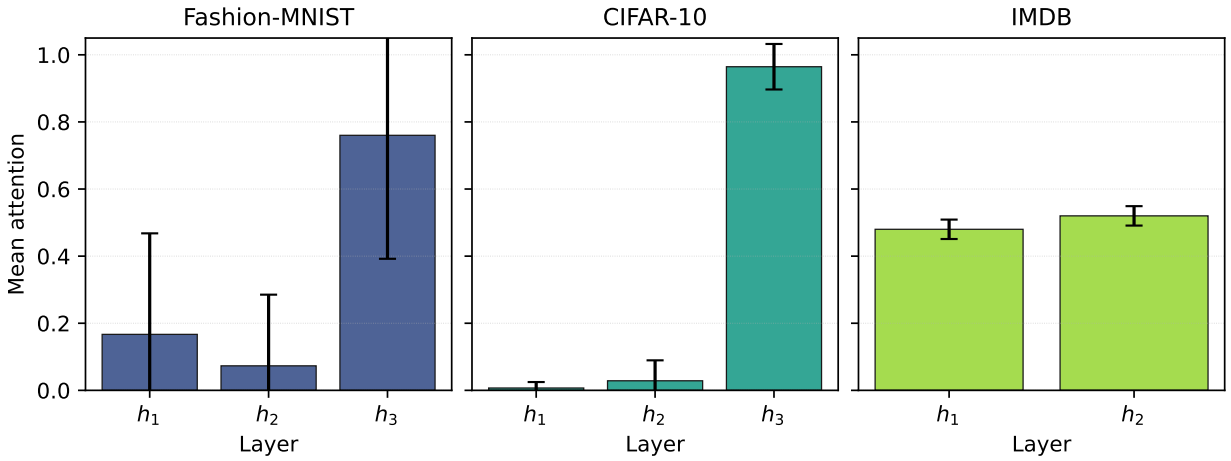

Figure 1: Global layer-wise attention statistics for LAYA (mean $\pm$ standard deviation over the full test set). LAYA consistently develops task-specific depth preferences: strong bias toward deep layers for CIFAR-10, softer depth usage for Fashion-MNIST, and nearly symmetric contributions in the IMDB text model.

On Fashion-MNIST, attention profiles are clearly class-dependent and markedly non-uniform. Most categories assign the largest share of attention to the deepest layer, but some systematically depart from this trend. For instance, class 5 (Sandal) concentrates most of its attention on the first layer in both the overall and correct subsets (mean ≈ 0.88 on h_1 for correct samples), while misclassified sandals shift part of the mass towards the deepest layer. A similar pattern is visible for class 0 (T-shirt/top): correct predictions favour the intermediate layer h_2, whereas incorrect ones move attention towards h_3. For class 9 (Ankle boot), both correct and incorrect predictions strongly rely on the last layer, but errors are associated with an even more extreme focus on h_3 and a reduced contribution from h_1. In contrast, categories such as Trouser (1), Pullover (2), Coat (4), Sneaker (7), and Bag (8) consistently prioritise the deepest layer, with only minor differences between correct and incorrect subsets. These patterns indicate that LAYA learns class-specific "attention fingerprints" over depth, and that shifts in where attention is concentrated correlate with success or failure on particular categories.

On CIFAR-10, attention becomes even more skewed: virtually all classes, irrespective of correctness, place almost all their mass on the deepest layer, with the first two layers receiving only residual weights. Differences between correct and incorrect predictions mainly translate into small redistributions among the earlier layers, confirming that high-level convolutional features dominate discrimination in this setting.

On IMDB, the behaviour is qualitatively different. With only two recurrent layers, attention remains nearly balanced between them for both sentiment classes and across correctness subsets, with mean values oscillating around 0.5 for h_1 and h_2. The heatmaps do not reveal strong class-specific signatures: the BiLSTM stack appears to maintain complementary information at both depths, and LAYA consistently treats the two layers as jointly relevant for classification.

Overall, the unified class-wise profiles confirm that LAYA adapts its use of depth to the structure of each domain: it develops distinct attention fingerprints per class in Fashion-MNIST, collapses onto the deepest representation in CIFAR-10, and adopts a balanced, two-layer strategy on IMDB.

Layer-wise attention in artwork classification. Figures 3 and 4 summarize the global and class-wise attention statistics of LAYA on the *Best Artworks of All Time* dataset. Taken together, these results indicate that LAYA does not treat depth in a purely monotonic, "deeper-is-better" fashion, but instead discovers a sparse, task-specific allocation of importance across ViT layers. The global profile in Fig. 3 shows that attention concentrates on a subset of intermediate and late layers (notably h_5, h_7, and h_9–h_{11}). In contrast, very shallow and very deep layers receive negligible mass. This pattern is consistent with the intuition that painterly style depends on mid-level cues such as texture, brushstroke statistics, and color composition, as well as higher-level structural cues, rather than on the most abstract semantics alone. The class-wise heatmaps in Fig. 4 further reveal distinctive "style fingerprints" in attention space: movements such as Byzantine Art and Proto Renaissance place substantial weight on earlier blocks, whereas Cubism, Neoplasticism, and Pop Art rely more heavily on later ones, reflecting their differing degrees of geometric abstraction and global structure. Correct predictions exhibit sharper, more characteristic attention patterns per style. At the same time, misclassified

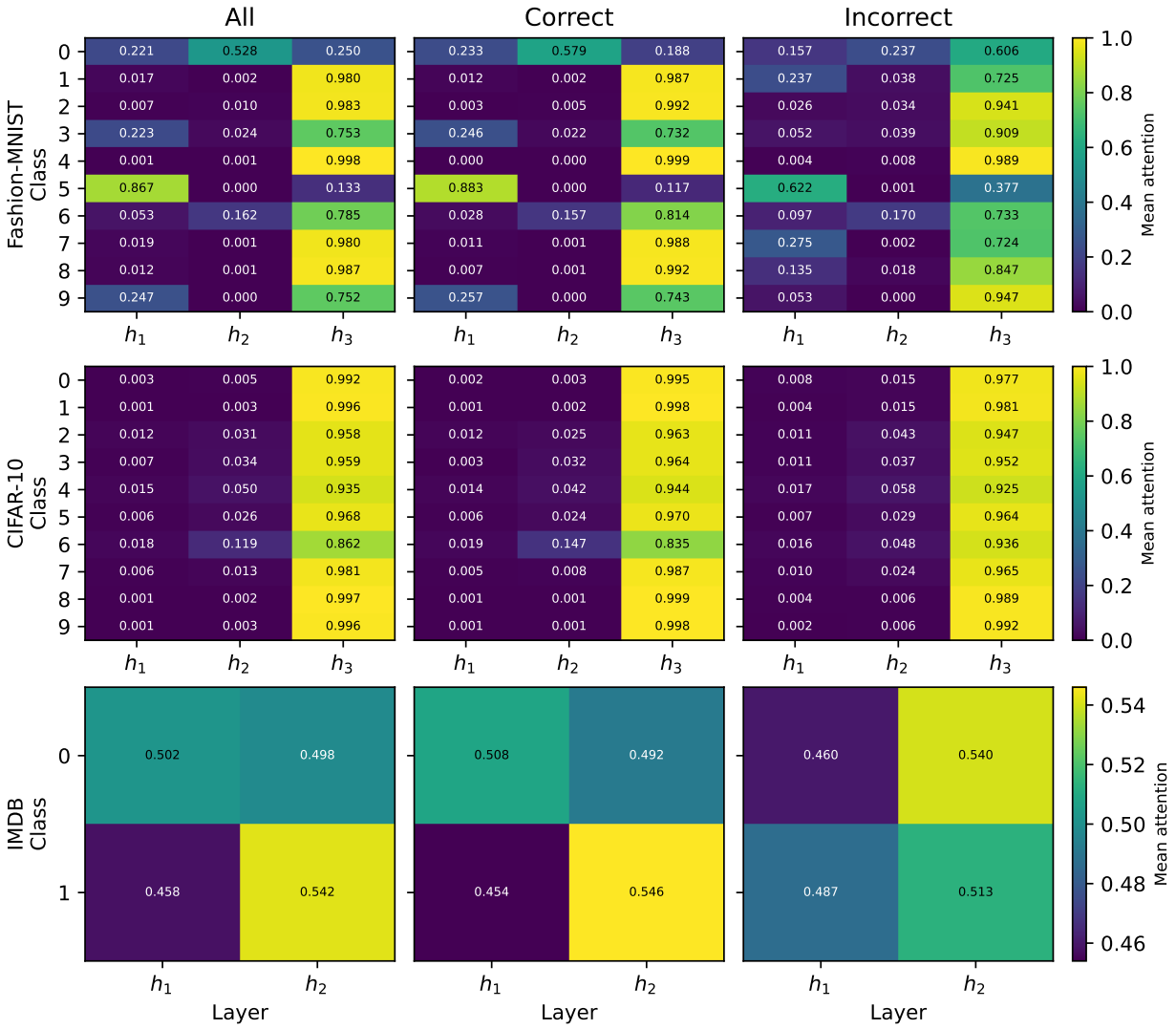

Figure 2: Class-wise mean attention profiles of LAYA across datasets. Rows correspond to Fashion-MNIST (top), CIFAR-10 (middle), and IMDB (bottom), while columns correspond to all test samples (left), correctly classified samples (center), and misclassified samples (right). Each cell reports the average attention $\bar{\alpha}_{c,i}$ assigned to layer i for class c.

samples tend to show more diffuse or shifted depth usage, suggesting that deviations from the canonical style-specific profile indicate uncertainty or confusion.

5 Conclusion and Future Work

This work revisited the conventional design choice of relying solely on the final hidden representation for prediction in deep networks. The proposed LAYA shows that the output stage can actively leverage the complete representational hierarchy by assigning input-dependent attention weights across depth. Experiments on both vision and language benchmarks indicate that this strategy leads to consistent improvements in predictive performance, while simultaneously providing intrinsic interpretability through layer-attribution scores that quantify the contribution of each abstraction level to the final decision.

Despite operating solely at the output stage, LAYA yields attention profiles that reveal clear and structured depth preferences. These profiles provide a principled basis for *attention-guided model compression*: layers that consistently